5508 module01

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This program (created 2026-01-28 by Reagan Todd) shows a brief introduction to the MIMIC III demo database and how to use the SELECT function as well as how to rename field names. This belongs in the public domain so please use however you see fit.

```
R.version.string
```

```
[1] "R version 4.5.2 (2025-10-31)"
```

```
Sys.Date()
```

```
[1] "2026-01-29"
```

```
library(tidyverse)
library(DBI)
library(RSQLite)
library(readr)
```

resources on the libaries

tidyverse is a package that contains various important libraries and by loading this libary, we are able to harness all of the core packages tidyverse offers (ggplot, tidyr, readr, etc.) without manually doing this. DBI stands for database interface, which allows us to connect databases within R. There is a great article on the specifics of DBI as the RSQLite.

creating the MIMIC III database from the demo folder

create a new SQLite database file

```
con <- dbConnect(SQLite(), "mimic_demo.db")

admissions <- read_csv(
  "~/Documents/Spring 26/MEDB-5530/data/mimic-iii-demo/ADMISSIONS.csv"
)
```

```
Rows: 129 Columns: 19
── Column specification ────────────────────────────────────────────────────────
Delimiter: ","
chr  (9): admission_type, admission_location, discharge_location, insurance,...
dbl  (5): row_id, subject_id, hadm_id, hospital_expire_flag, has_chartevents...
dttm (5): admittime, dischtime, deathtime, edregtime, edouttime
```

```
ℹ Use `spec()` to retrieve the full column specification for this data.
ℹ Specify the column types or set `show_col_types = FALSE` to quiet this message.
```

```
dbWriteTable(con, "admissions", admissions, overwrite = TRUE)
```

explanation of functions used

dbWriteTable is a way to take a csv file and translate it into a database form that R will recognize. If you do not have a database, much like the downloaded demo for the MIMIC III data, this is required. Please note that this just translated a single data table (admissions) from the MIMIC III data.

Q1. Read the admissions table and display the first ten records and the following fields.

–Admission_type (self-explanatory) –Admission_location (self-explanatory) –Discharge_location (self-explanatory) –Insurance (self-explanatory)

```
cox <- dbConnect(SQLite(), "mimic_demo.db")
admissions <- dbGetQuery(conn=cox, "
    select
      admission_type,
      admission_location,
      discharge_location,
      insurance
    from admissions
    LIMIT 10;
")
admissions
```

```
   admission_type        admission_location       discharge_location insurance
1       EMERGENCY      EMERGENCY ROOM ADMIT         HOME HEALTH CARE  Medicare
2       EMERGENCY TRANSFER FROM HOSP/EXTRAM             DEAD/EXPIRED   Private
3       EMERGENCY TRANSFER FROM HOSP/EXTRAM             DEAD/EXPIRED  Medicare
4       EMERGENCY      EMERGENCY ROOM ADMIT                      SNF  Medicare
5       EMERGENCY TRANSFER FROM HOSP/EXTRAM             DEAD/EXPIRED  Medicare
6       EMERGENCY      EMERGENCY ROOM ADMIT REHAB/DISTINCT PART HOSP  Medicare
7        ELECTIVE PHYS REFERRAL/NORMAL DELI                      SNF  Medicare
8       EMERGENCY      EMERGENCY ROOM ADMIT                      SNF  Medicare
9       EMERGENCY      EMERGENCY ROOM ADMIT REHAB/DISTINCT PART HOSP  Medicare
10      EMERGENCY      EMERGENCY ROOM ADMIT                      SNF  Medicare
```

```
dbDisconnect(cox)
```

interpretation of output

The above table shows the Admissions data for the first 10 patients. In this table, only the selected fields of admission type, admission location, discharge location, and insurance are shown. There are 10 rows and 4 columns.

explanation of functions used

The select function used in SQL retrieves the rows from the database. The select function must be used first, and essentailly tells the query what to find. If an * is used, all columns/rows will be returned. If select is followed by field names, such as in Q1, then it will only return the listed columns.

The from function is used with the select function to tell the query where the data can be found. It is generally listed in the format of FROM database.datatable.

Q2. Read the admissions table and rename the column insurance to admission_insurance. Display the first ten records.

```
cox <- dbConnect(SQLite(), "mimic_demo.db")
admissions <- dbGetQuery(conn=cox, "
    select
      insurance as admission_insurance
    from admissions
    LIMIT 10;
")
admissions
```

```
   admission_insurance
1            Medicare
2             Private
3            Medicare
4            Medicare
5            Medicare
6            Medicare
7            Medicare
8            Medicare
9            Medicare
10           Medicare
```

```
dbDisconnect(cox)
```

interpreting the output

Above is the table showing the renaming of the field name Insurance to admission_insurance. There are 10 rows and 1 column.

explantation of keywords used

The as keyword is used to change the indicated field name. This can be used to rename/change multiple field names in the same chunk.